

Epidemiology of soy exposures and breast cancer risk

Most of the early studies published on soy and breast cancer were not designed to test the effect of soy; the assessment of soy intake was usually crude and few potential confounders were considered in the analysis. In this review, we focused on studies with relatively complete assessment of dietary soy exposure in the targeted populations and appropriate consideration for potential confounders in the statistical analysis of study data. Meta-analysis of the 8 (1 cohort, 7 case-control) studies conducted in high-soy-consuming Asians show a significant trend of decreasing risk with increasing soy food intake. Compared to the lowest level of soy food intake (■5 mg isoflavones per day), risk was intermediate (OR=0.88, 95% confidence interval (CI)=0.78–0.98) among those with modest (~10 mg isoflavones per day) intake and lowest (OR=0.71, 95% CI=0.60–0.85) among those with high intake (■20 mg isoflavones per day). In contrast, soy intake was unrelated to breast cancer risk in studies conducted in the 11 low-soy-consuming Western populations whose average highest and lowest soy isoflavone intake levels were around 0.8 and 0.15 mg per day, respectively. Thus, the evidence to date, based largely on case-control studies, suggest that soy food intake in the amount consumed in Asian populations may have protective effects against breast cancer.